

ICT 305

Computer Networks & Data Communication

Simplified Study Notes

1. Data Transmission, Layers and Protocols
2. How the Internet Works
3. Internet Architecture
4. Network Models
5. The OSI Model
6. The Application Layer
7. Forms of Network Addresses

1. Data Transmission, Layers and Protocols

When you send a message over a network (for example, a WhatsApp text or an email), the data does not just "fly" from your phone to your friend's phone in one piece. It is broken down, processed, sent in stages, and rebuilt at the other end. To make this organised and predictable, computer scientists group the work into **layers**, and each layer follows agreed-upon rules called **protocols**.

What is Data Transmission?

Data transmission is simply the process of sending data (in the form of electrical signals, light pulses, or radio waves) from one device to another over a communication medium such as a cable, fibre optic line, or the air (wireless).

- **Analog transmission** – data is sent as a continuous wave (e.g., old telephone lines).
- **Digital transmission** – data is sent as discrete signals, usually 0s and 1s (this is how almost all modern networks work).
- **Simplex** – data flows in one direction only (e.g., a TV broadcast).
- **Half-duplex** – data flows in both directions, but not at the same time (e.g., a walkie-talkie).
- **Full-duplex** – data flows in both directions at the same time (e.g., a phone call).

Why Do We Need Layers?

Imagine if one single piece of software had to handle everything: turning your typed message into electrical signals, finding the destination address, making sure no data was lost, AND displaying it nicely on the screen. That would be extremely complicated and hard to fix when something goes wrong.

Instead, network designers split the work into **layers**. Each layer has one specific job, and it only needs to talk to the layer directly above and below it. This is called a **layered architecture**.

Remember: Think of layers like a company. The cleaner doesn't need to know how to do accounting, and the accountant doesn't need to know how to fix the printer. Each person (layer) has a clear, separate responsibility, but they all work together to keep the company (network) running.

What is a Protocol?

A **protocol** is a set of rules that devices agree to follow so they can understand each other. Just like two people need to speak the same language to have a conversation, two computers need to use the same protocol to exchange data correctly.

- **TCP (Transmission Control Protocol)** – breaks data into packets, makes sure all packets arrive, and puts them back in order. Reliable but slightly slower.
- **IP (Internet Protocol)** – gives every device an address and decides the best path for data to travel.
- **HTTP/HTTPS** – rules for loading websites in a browser.
- **FTP** – rules for transferring files between computers.
- **SMTP** – rules for sending email.
- **DNS** – rules for translating website names (like google.com) into IP addresses.

Example: When you open a website, your browser and the web server agree to use the HTTP protocol. The browser asks ("GET me this page"), and the server replies in a format the browser understands. If they used different rules, the page would never load.

2. How the Internet Works

The Internet is simply a massive global network of networks. It connects billions of devices — phones, laptops, servers, routers — so they can exchange data. No single company or government owns the whole Internet; instead, many networks agree to connect to each other and follow the same set of protocols (mainly TCP/IP).

Step-by-Step: What Happens When You Visit a Website

- **Step 1 – You type an address.** You type "www.example.com" into your browser.
- **Step 2 – DNS lookup.** Your computer doesn't understand names, only numbers (IP addresses). So it asks a DNS server: "What is the IP address for example.com?"
- **Step 3 – Data is broken into packets.** Your request is broken into small chunks called packets, each labelled with the destination address.
- **Step 4 – Routing.** The packets travel through your home router, then your Internet Service Provider (ISP), then through many other routers across cities, countries, or undersea cables, until they reach the destination server.
- **Step 5 – The server responds.** The web server receives the request, prepares the webpage, and sends it back — again broken into packets.
- **Step 6 – Reassembly.** Your computer collects all the packets (which may have travelled different paths) and reassembles them into the complete webpage you see.

Remember: All of this can happen in well under one second, even though the data might travel thousands of kilometres and pass through dozens of routers!

Key Components That Make the Internet Work

- **ISPs (Internet Service Providers)** – companies that connect homes and businesses to the wider Internet.
- **Routers** – devices that decide the best path to send each packet toward its destination.
- **Servers** – powerful computers that store websites, files, and services, and respond to requests.
- **Undersea/fibre cables and satellites** – the physical pathways that carry data between countries and continents.
- **IP addresses** – unique numbers that identify every device, so data knows where to go.

3. Internet Architecture

Internet architecture describes **how the Internet is structured** — the design and arrangement of the networks, hardware, and protocols that allow global communication. It explains the "shape" of the Internet, not just how one website loads.

The Client–Server Model

Most of the Internet works on a **client–server** relationship:

- **Client** – the device requesting a service (e.g., your phone's browser).
- **Server** – the device providing the service (e.g., a web server hosting a website).

Example: Your phone (client) sends a request to YouTube's servers, and the servers send back the video you want to watch.

Hierarchical Structure of the Internet

The Internet is organised in layers of networks, often described as Tiers:

- **Tier 1 networks** – the giant backbone providers that own long-distance fibre cables spanning continents and oceans. They connect directly to each other.
- **Tier 2 networks** – regional or national ISPs that buy access from Tier 1 networks and also connect to some Tier 2s directly.
- **Tier 3 networks** – local ISPs that connect directly to homes, schools, and businesses (this is usually the ISP you personally pay).

Decentralisation: No Single Point of Control

One of the most important architectural principles of the Internet is that it is **decentralised** and **packet-switched**. There is no single central computer controlling everything. If one path or router fails, data can be re-routed through another path. This makes the Internet resilient — a key design goal from its very beginning (it was originally designed to survive partial damage to the network).

Packet Switching vs Circuit Switching

- **Circuit switching** – a dedicated path is reserved for the entire communication (like old telephone calls). Reliable but wastes capacity when idle.
- **Packet switching** – data is broken into packets, and each packet can take a different path to the destination, then gets reassembled. This is what the Internet uses — it is more efficient and resilient.

4. Network Models

A **network model** is a conceptual framework (a blueprint) that describes how data communication should be organised into layers, with each layer responsible for a specific task. Network models help engineers design, build, and troubleshoot networks in a structured, standardised way — instead of everyone inventing their own random system.

Why Use a Model at All?

- It breaks a complex process (networking) into smaller, manageable pieces.
- It allows different companies to build hardware/software that all work together, as long as they follow the same model.
- It makes troubleshooting easier — you can isolate which layer has a problem.
- Changes in one layer (e.g., upgrading cables) don't require rebuilding every other layer.

The Two Major Network Models

- **OSI Model (Open Systems Interconnection)** – a theoretical, 7-layer model created by ISO to standardise networking concepts. It is mostly used for teaching and as a reference.
- **TCP/IP Model** – the practical, 4-layer model that the actual Internet is built on (Application, Transport, Internet, Network Access/Link).

We will look at the OSI Model in detail in the next section, since it is the most commonly examined model.

5. The OSI Model

The **OSI Model (Open Systems Interconnection Model)** is a 7-layer conceptual framework that standardises how different computer systems communicate over a network. It was developed by the International Organization for Standardization (ISO). Each layer has a specific job, and data passes down through the layers on the sending side, then back up through the layers on the receiving side.

Layer	Name	Main Job	Simple Example
7	Application	Interface between the user and the network	Browser, email app, WhatsApp
6	Presentation	Formats, encrypts, and compresses data	SSL encryption, JPEG, file compression
5	Session	Opens, manages, and closes communication sessions	session staying active on a website
4	Transport	Ensures complete, reliable data delivery	TCP (reliable), UDP (fast, no guarantee)
3	Network	Handles addressing and routing between networks	IP addresses, routers
2	Data Link	Handles addressing on a local network and error checking	MAC address, switches
1	Physical	Transmits raw bits as electrical/light/radio signals	Cables, fibre optics, Wi-Fi signals

Remember: An easy way to remember the order from Layer 7 down to Layer 1 is the phrase: *"All People Seem To Need Data Processing"* (Application, Presentation, Session, Transport, Network, Data Link, Physical).

How Data Moves Through the Layers (Encapsulation)

When you send data, it travels **DOWN** the layers on your device (from Application to Physical). At each layer, a small piece of control information (called a **header**) is added to the data. This process is called **encapsulation**. When the data arrives at the destination, it travels **UP** the layers (from Physical to Application), and each layer removes its corresponding header. This reverse process is called **decapsulation**.

Example: Imagine packing a gift inside a box, then that box inside another box, then inside a shipping crate (encapsulation). When it arrives, the receiver opens the crate, then the box, then finally the gift (decapsulation). Each layer added its own 'wrapping' for a specific purpose.

6. The Application Layer

The **Application Layer** is Layer 7 — the topmost layer of the OSI model, and the one closest to the end user. It is NOT the application itself (like your browser or email app); rather, it provides the network services and protocols that applications use to communicate with each other over a network.

What Does the Application Layer Actually Do?

- Provides the rules (protocols) that let applications request and exchange data over a network.
- Identifies the communicating partners (who you are talking to) and checks resource availability.
- Handles things like file transfers, email delivery, web browsing, and remote login.
- It is the only layer that directly interacts with the actual application software used by the human user.

Common Application Layer Protocols

Protocol	Used For
HTTP / HTTPS	Loading websites (HTTPS is the secure, encrypted version)
FTP	Transferring files between computers
SMTP	Sending email
POP3 / IMAP	Receiving and managing email
DNS	Translating domain names (google.com) into IP addresses
DHCP	Automatically assigning IP addresses to devices on a network
Telnet / SSH	Remotely logging into and controlling another computer

Remember: A common mix-up: "the Application Layer" is not the same as "an application" like Instagram. Instagram (the app) USES Application Layer protocols (like HTTPS) behind the scenes to fetch your photos and messages over the network.

7. Forms of Network Address

For any device to send or receive data on a network, it needs a unique identifier — an **address** — so that other devices know exactly who to send data to and where replies should go. There are several different forms of network addresses, each used at a different layer or for a different purpose.

1. MAC Address (Physical / Hardware Address)

- Operates at the **Data Link Layer** (Layer 2).
- A unique 48-bit address burned into the network card by the manufacturer (e.g., 00:1A:2B:3C:4D:5E).
- Used for communication within the same local network (LAN).
- It does NOT change, even if you move the device to a different network (unlike an IP address).

2. IP Address (Logical Address)

Operates at the **Network Layer** (Layer 3). It identifies a device's location on a network so data can be routed to it from anywhere, including across the Internet.

- **IPv4** – written as four numbers separated by dots, e.g., 192.168.1.10. Provides about 4.3 billion addresses, which is now running out.
- **IPv6** – written in hexadecimal groups separated by colons, e.g., 2001:0db8:85a3::8a2e:0370:7334. Provides an enormously larger address space to support the growing number of devices.
- **Public IP address** – a unique address visible on the wider Internet (assigned by your ISP).
- **Private IP address** – used only within a local network (e.g., your home Wi-Fi), such as 192.168.x.x or 10.x.x.x.

3. Port Address

Operates at the **Transport Layer** (Layer 4). While an IP address identifies the device, a port number identifies the specific **application or service** running on that device.

Example: A web server might use port 80 (HTTP) or port 443 (HTTPS), while an email server uses port 25 (SMTP). Your computer can run many services at once, all sharing the same IP address but using different port numbers.

4. Domain Name / URL

This isn't a "raw" network address, but a human-friendly name (like `www.example.com`) that DNS translates into the actual IP address needed for routing. It exists because remembering names is far easier for humans than remembering long strings of numbers.

Quick Comparison Table

Address Type	OSI Layer	Identifies	Example
MAC Address	Data Link (2)	Physical network device	00:1A:2B:3C:4D:5E

IP Address	Network (3)	Device location on a network	192.168.1.10
Port Number	Transport (4)	Specific application/service	Port 443 (HTTPS)
Domain Name	Application (7)	Human-readable site name	www.example.com

Remember: To send a letter, you need a house address (IP address) AND a person's name at that house (port number) — the postal worker (router) only cares about the house address, but the right family member (application) only receives it once it's handed over correctly inside the house.

— End of Notes —